ADRIAN CORTLANDT

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EDUCATION

University of Florida Gainesville, FL
B.S. Computer Engineering | B.A. Economics *Expected May 2027*
GPA: 3.6/4.0

Relevant Coursework: Real-Time Operating Systems, Autonomous Robotics, Microprocessor Applications, Digital Logic, Data Structures and Algorithms, Computer Organization, Digital Design, Circuits 1.

Study Abroad: Nanyang Technological University, Singapore - Data Science & AI program (June 2025 - July 2025)

Certifications: Microsoft Certified: AI-900 (Azure AI), SC-900 (Security & Compliance), AZ-900 (Azure Fundamentals).

TECHNICAL SKILLS

Languages: C/C++, C#, Python, VHDL, SystemVerilog, ARM Assembly, R.
Embedded Systems: Microcontrollers (ARM/AVR), bare-metal firmware, device drivers, board bring-up, RTOS development, ROS 2 (Jazzy), UART/SPI/I2C/RS232, ADC/DAC, DMA.
FPGA & Hardware: FPGA design, RTL design, Questa/ModelSim, Quartus, timing analysis, waveform debugging, LTspice.
Test & Instrumentation: Oscilloscope, logic analyzer, function generator, multimeter, Digilent Analog Discovery (DAD), unit testing.
Tools & Frameworks: Git/GitHub/GitLab, Jira, XML, Unreal Engine, Linux CLI, WSL, Visual Studio, Atmel Studio, MATLAB.

PROFESSIONAL EXPERIENCE

Lockheed Martin - Rotary and Mission Systems **May 2026 – Present**
Software Engineering Intern *Orlando, FL*

- Developing an enterprise test-automation platform replacing legacy software used across Lockheed Martin.
- Built a C# serial terminal to capture and decode 12 SPI message types from the test station at 19200 baud over COM1.
- Migrated legacy COM communication to gRPC to modernize inter-service and software-to-hardware messaging.
- Extended the test-execution engine's API layer to support configurable test properties, serializing to and from a defined XML schema.
- Worked within an Agile development cycle in Jira, contributing to sprint planning, daily scrums, and code reviews.

Integration Lead - Intern Project Competition | **1st Place**

- Elected by a 20-person intern cohort to lead the integration team, based on prior embedded systems experience.
- Delivered a counter-drone training simulator in an 8-week timeline, from a wooden post, an Xbox controller, and a VN-100 IMU.
- Owned the VN-100 end to end, capturing and decoding its output, calibrating it, and streaming its orientation data into Unreal.
- Presented the completed system and supporting business case to Lockheed Martin executives, VPs, and program management.

DriCloud **June 2025 – August 2025**
Software Engineering Intern *Virtual*

- Tested embedded software integrations on Microsoft Azure, validating third-party API connectivity.
- Collaborated with senior engineers to diagnose software synchronization issues, increasing system reliability.
- Automated data transfer scripts that reduced processing time by 20% and improved workflow efficiency.

Gator Motorsports - Formula SAE **August 2024 – May 2025**
Electrical Systems Team Member *Gainesville, FL*

- Prototyped and tested electrical subsystems for a Formula race car, improving sensor reliability and cutting system noise by 15%.
- Led analog sensor integration and collaborated across software and mechanical subteams to ensure stable data acquisition under racing conditions.
- Mentored newer members on circuit debugging and documentation practices for future iterations.

PROJECTS

Embedded Peripheral Driver Suite - ATxmega128A1U, C

- Wrote bare-metal drivers for SPI, UART, ADC, DAC, DMA, timer, and event-system peripherals through register configuration.
- Debugged and optimized interrupt-driven data acquisition and waveform generation at 32 MHz.

Custom 8-Bit CPU - VHDL, ModelSim

- Designed and implemented an 8-bit microprocessor in VHDL with a custom ALU, operand registers, and a control FSM decoding a 7-instruction ISA.
- Validated timing and logical correctness through waveform analysis in ModelSim, verifying instruction decoding and hardware synchronization.

Real-Time Face Detection Node - ROS 2, OpenCV, Python

- Engineered a real-time ROS 2 facial recognition node in OpenCV, using a publish/subscribe architecture to manage asynchronous data streams and sensor topics.
- Bridged physical cameras to Ubuntu WSL2 via usbipd, verifying low-latency performance in rqt_image_view.